Class-Adaptive Data Augmentation for Road Damage Detection

Index Terms—road damage detection, data augmentation, class-adaptive learning, augmentation policy, object detection

I. Introduction

Accurate detection of pavement defects are important components of roadway condition assessment, providing information needed to evaluate pavement deterioration and guide maintenance decisions. Traditionally, these assessments have relied heavily on manual inspections, which can be labor-intensive and difficult to implement consistently across extensive road networks. Advances in deep learning have created opportunities to automate this process, with promising results in pavement damage detection and classification [1]–[3]. Despite this progress, achieving reliable performance under varying roadway and image acquisition conditions remains challenging. Differences in illumination, pavement texture, camera perspective, and defect appearance introduce considerable visual variability that may not be adequately represented during model training. This challenge is further complicated by the visual similarities among pavement damage categories, where distinctions often depend on specific geometric and structural characteristics, such as crack orientation, connectivity, and spatial distribution. Consequently, improving model generalization requires accounting for variations in pavement appearance while preserving the visual features that distinguish individual damage categories throughout the image.

Data augmentation offers a practical means of addressing this challenge by introducing controlled variations into training images, allowing models to learn from a broader range of visual conditions without requiring additional data collection. However, the effectiveness of this approach depends not only on the diversity of the transformations applied but also on their ability to preserve the characteristics that define each damage category. For example, geometric transformations may alter the directional features separating longitudinal and transverse cracks, while excessive blurring or occlusion can obscure the fine structural details of interconnected cracks. These effects suggest that the same transformation may provide useful variability for one damage category while compromising essential visual information in another. In detection, this tension arises within a single training image rather than only between images: standard pipelines typically transform the whole frame, so every damage it contains receives the same transformation. As such, a transformation that helps the detector recognize one damage category may degrade another category in the same image, and augmentation policies applied uniformly across all classes may introduce variations that are beneficial to some categories but detrimental to others. Such differences may also be overlooked when augmentation effectiveness is assessed primarily through aggregate performance, as improvements in aggregate accuracy can coexist with reduced performance for individual classes [4]. This motivates the need for augmentation strategies that account for class-specific sensitivities to different image transformations and can act on each class within a multi-class image.

Existing augmentation methods have progressively addressed this limitation. AutoAugment and RandAugment optimize or control augmentation policies primarily through shared transformation strategies [5], [6], and detection-specific extensions add operations that act on annotated boxes or adapt to object scale [7], [8], but their policies are not conditioned on object category. Class-aware approaches recognize that different categories may require distinct augmentation policies [9]–[11], and road-damage detection work has varied augmentation by damage type through probabilities set before training [12]. Nevertheless, policies determined before training cannot directly respond to changes in model sensitivity as learning progresses. Online approaches, including CTAugment and CUDA, address this issue by adapting augmentation using feedback from the evolving model [13], [14]. However, CTAugment pools feedback across classes, while CUDA maintains a class-level severity state; both are formulated for classification, where each training sample carries one label, so neither specifies which class's state should govern an image that contains several. This distinction matters because a detector may tolerate strong blurring for one damage class while another, thinner class loses the structure that identifies it. A single state applied to the whole image forces a compromise between them, potentially imposing unnecessary restrictions on beneficial transformations for one class or excessive distortion on the other.

These limitations motivate an augmentation framework that accounts for damage category, the detector's evolving response, and the region being transformed. The proposed framework groups connected overlapping boxes and applies one consistent edit to each selected group. Paired reference and transformed images provide class-specific feedback for photometric and blur/noise

operations. Detection retention constrains strength for every class represented in the group, while transformation-induced loss guides family selection. Separate groups can therefore follow different policies, while classes sharing pixels retain a joint constraint.

The central hypothesis is that online class-conditioned control, applied to local groups of damage boxes, improves overall and per-class detection relative to shared augmentation policies. This hypothesis requires evaluation on unaugmented images, including scenes with multiple and overlapping damage classes. To investigate it, we propose the following framework, mechanism, and evaluation protocol:

1) An online class-adaptive augmentation framework that regulates transformation strength and family selection by damage category and applies them locally in multi-class road-damage images, with joint tolerance constraints for overlapping boxes.
2) A feedback-driven adaptation mechanism combining per-class detection retention for strength regulation with transformation-induced loss for family selection.
3) A controlled evaluation protocol on RDD2022 to characterize damage co-occurrence and class-dependent augmentation sensitivity, separate the effects of local application and class conditioning, and assess detection performance and computational overhead.

II. Related Work

The use of data augmentation in pavement damage recognition has developed around the need to obtain informative training examples from limited annotated images. Road-specific studies have approached this need through both the choice of transformations and the construction of additional samples. Tsuchiya et al. [9] compared augmentation methods by their influence on class-level results and used the selected methods to expand the training data. Aghayan-Mashhady and Amirkhani [15] investigated bounding-box augmentation and generative adversarial synthesis, examining the effect of unrealistic examples on road-damage detection. Detection systems built on multi-country road-damage data tie augmentation more directly to damage types and instances: one Detectron2 and Faster R-CNN system set augmentation probabilities for each damage type and country from data exploration and tested GAN-generated damage patches [12], while a content- and perspective-aware cut-and-paste method accounts for road location and perspective when placing damage instances [16]. Their findings motivate examining augmentation through both the variation it introduces and the damage information retained in the transformed image. The design of the augmentation policy therefore matters alongside the number of examples it produces.

Choosing suitable transformations becomes more difficult as the number of operations, application probabilities, and magnitudes increases. Automatic augmentation methods address this problem at different levels of complexity. AutoAugment [5] searches combinations of these variables using validation performance, whereas RandAugment [6] retains a smaller search space comprising operation count and a common magnitude. TrivialAugment [17] simplifies the procedure further by randomly sampling a single operation and its strength. This progression demonstrates that useful augmentation need not depend on an elaborate search procedure. It also establishes shared policies as an important reference point: transformations are selected according to a common rule, leaving open how their effects are distributed across the classes to which that rule is applied.

Object detection adds a decision that classification does not have: a transformation can act on the whole image or only on individual annotated boxes. Zoph et al. [7] learn augmentation policies for detection, including operations restricted to box content, and Scale-Aware AutoAug [8] searches scale-aware policies at both the image and box levels. InterAug [18] provides a tuning-free augmentation policy for data-efficient and robust detection. These methods show that box-level application is feasible within detection pipelines. Their policies, however, are chosen for the dataset or for object scale rather than for each category, so they leave open how a transformation should differ between classes that appear in the same image.

Those class-level effects are consequential even when the shared policy improves overall accuracy. Kirichenko et al. [4] found that augmentation gains on ImageNet can coexist with substantial losses for individual categories, including categories affected by fine-grained distinctions and label ambiguity. This finding complements the class-dependent results reported for road damage by Tsuchiya et al. [9]: a transformation's value cannot be inferred solely from its average effect across the dataset. Class-conditioned policy learning incorporates this distinction directly. Yoo and Kang [10] derive class-specific transformation parameters from an initially trained classifier, then use the parameters with scheduled decay during retraining. AdaAug [11] learns policies conditioned on class and potentially on individual instances, which are subsequently reused for classifier training. These approaches establish the relevance of conditioning augmentation on the class of the example being transformed. Because they were developed for classification, that conditioning is defined per sample, with one label selecting the policy; an image containing several classes has no single label to condition on. A separate question is when the policy should change: an offline selection or predetermined schedule does not use feedback from the model's changing response during the subsequent training run.

Online augmentation links that response to policy updates, allowing the training challenge to evolve with the learner. CTAugment, introduced within ReMixMatch [13], uses predictions on transformed labeled examples to up-

date magnitude-sampling statistics for individual operations. CUDA [14] instead evaluates augmented examples within each class and adjusts a class-specific severity state, which determines operation count and magnitude. The two mechanisms retain different information from the same general source of feedback. CTAugment distinguishes transformations while pooling observations across classes; CUDA distinguishes classes while controlling transformation strengths through a common state within each class. Their comparison brings the representation of model tolerance into focus: adaptation depends not only on obtaining feedback, but also on which differences that feedback is allowed to express, and in multi-class detection images, on which region of the image that feedback is attributed to.

Learned policy networks provide a more expressive way to represent such differences. MADAug [19] jointly optimizes the classifier and an input-dependent augmentation network, using a bilevel objective to learn operation probabilities and magnitudes as training progresses. AdaAugment [20] uses actor–critic learning and classification-loss feedback to adapt instance-level magnitudes. In these formulations, the augmentation policy becomes a learned component of the training process, enabling decisions to vary with individual inputs and the evolving model. This approach differs from direct-feedback controllers in where adaptation is represented: a policy network learns how to assign transformations, whereas a state-based controller explicitly records the model's response to specified augmentation conditions. The comparison therefore concerns the mechanism and granularity of adaptation, as well as whether adaptation occurs online.

The interpretation of model feedback is equally important to the choice of policy representation. A transformation can increase loss by exposing a useful variation, but it can also do so by obscuring information needed to recognize the original category. TeachAugment [21] addresses this ambiguity by seeking transformations that challenge a target model while remaining recognizable to a teacher. CUDA's performance-based strength adjustment and CTAugment's tolerance estimates likewise connect augmentation difficulty to the model's ability to recognize transformed examples [13], [14]. These approaches support treating difficulty and recognizability as related but distinct criteria. For pavement defects, that distinction matters when geometric distortion alters directional cues or when blur and occlusion conceal crack structure: a harder example is not necessarily a better training example. Detection retention can provide evidence of current model tolerance, but cannot by itself establish semantic label preservation.

The resulting limitation is most apparent when class sensitivity and spatial application must be considered together. In the direct-feedback formulations represented by CTAugment and CUDA, a favorable response pooled across classes can conceal a sensitive category, while a class-level severity state has no defined owner in an image containing several classes. Shared detection policies [7], [8] can act on boxes but do not distinguish between categories, and class-dependent road-damage practice sets its probabilities before training [12]. For example, an image-level transformation chosen to suit a pothole may erase a thin longitudinal crack in the same frame, even if the detector tolerates the transformation on potholes. Road-specific studies establish the importance of class-dependent augmentation, while online and learned policies show how augmentation can respond to learning progress. What remains to be established is whether directly measuring each class's tolerance and applying the result at the box level provides effective control in multi-class images without learning an additional policy network.

This study therefore investigates whether direct class-level feedback and local application can reduce the compromise imposed by image-level augmentation in multi-class scenes. Separate groups of damage boxes can receive different policies; overlapping boxes receive a shared edit constrained by the measured retention of every represented class. Loss feedback then guides selection among eligible transformation families. Controlled comparisons will test local application, class conditioning, online updates, family-specific strength, and difficulty-based selection. Table I positions this proposed mechanism relative to the reviewed approaches.

III. Methodology

A. Problem Formulation and Overview

We study augmentation for full-image road-damage detection. Each training image x has annotations $Y = \{(b_j, c_j)\}_{j=1}^{n}$, where b_j is a bounding box and c_j is one of four RDD2022 categories: longitudinal cracks, transverse cracks, alligator cracks, and potholes [22]. A detector f_θ predicts boxes and classes from the complete image. Ground-truth boxes guide augmentation during training; inference uses only the trained detector.

The controller chooses a transformation family and strength for a local group of boxes. Separate groups can follow different class-conditioned policies, but overlapping boxes cannot receive independent edits at shared pixels. We therefore use a common edit within each connected group, subject to a retention criterion for every represented class. This localizes the shared decision without eliminating the trade-off within the group. Detection improvements remain a hypothesis to be tested.

B. Overall Framework

Training alternates between detector optimization and policy updates (Fig. 1). After a common warm-up, the frozen detector evaluates paired reference and transformed training images. Class-specific retention determines the eligible strengths; normalized loss increases determine how often eligible families are sampled, with a uniform component for exploration. The policy supplies one local

TABLE I
Augmentation Mechanisms and Their Relationship to the Proposed Framework

Study	Application setting	Policy and feedback	Limitation or distinction relative to this study
Tsuchiya et al. [9]	Road-damage recognition	Offline class-specific method selection using model performance	Selection is not updated from the final model's changing response.
Detectron2 system [12]	Multi-country road-damage detection	Fixed augmentation probabilities per damage type and country, set from data exploration	Probabilities are set before training and do not use feedback from the detector.
Zoph et al. [7]	General object detection	Learned policies, including operations restricted to box content	Policy is chosen for the dataset, not for each category.
Scale-Aware AutoAug [8]	General object detection	Scale-aware policies searched at image and box levels	Conditioned on object scale rather than on category.
Yoo and Kang [10]	Digit and wafer-map classification	Class-specific parameter search followed by scheduled decay during retraining	The retraining schedule does not respond to current model tolerance.
AdaAug [11]	General and fine-grained recognition	Learned class- and instance-conditioned policies reused for classifier training	Conditioning is defined per single-label sample; the present study examines direct feedback during final training.
CTAugment [13]	Semi-supervised classification	Online updates of operation-specific magnitude statistics	Feedback pooled across classes does not retain class-specific tolerance.
CUDA [14]	Long-tailed recognition	Online class-level severity controlling operation count and magnitude	Class-level state is defined for single-label samples and has no defined owner in a multi-class image; operations are selected uniformly.
MADAug [19]	General image classification	Joint policy/classifier optimization of input-dependent operations and magnitudes	Establishes broader online control through a learned policy; explicit class-level probing is a different mechanism.
AdaAugment [20]	General and long-tailed recognition	Actor–critic updates of instance-level augmentation magnitude	Learns instance magnitudes rather than maintaining explicit class-level tolerance states.
Proposed framework	Road-damage detection in multi-class images	Paired class-level feedback controls strength and family selection for local overlap groups	Tests class-conditioned local control with joint constraints on overlapping boxes; detection benefit and cost remain to be evaluated.

CTAugment is the augmentation component of ReMixMatch. MADAug also uses a separate augmentation-probability schedule.

edit per augmented training image and is refreshed at fixed intervals as the detector changes.

Policies are indexed by a group's class set s. A group containing only class c uses $s = \{c\}$; mixed groups use all represented classes, giving at most 15 nonempty sets for four categories. Probes pool feedback across groups with the same set while retaining a separate estimate for each class. This defines a compact controller without a learned policy network, but it estimates average responses across sampled groups, not the tolerance of each individual group. Group size, geometry, and acquisition source may still affect those responses.

C. Overlap Groups and Local Transformations

Boxes are clipped and rasterized to the image grid before grouping, using the same support convention as the edit masks. They form graph nodes, with edges between boxes sharing positive area. Connected components define the overlap groups; an isolated box forms its own group. This criterion concerns box support, not overlap of the underlying defect shapes. For a selected group G, let M_G be its union mask. Operation o at discrete strength m produces

$$\widetilde{x} = x + \alpha_G \odot \big(T_{o,m}(x) - x\big), \tag{1}$$

where $\odot$ denotes pixelwise multiplication and the mask α_G has values in $[0, 1]$, is zero outside M_G, and tapers inward at its boundary. The transform is computed from the reference image and composited once across the union. Shared pixels are therefore edited once, independently of box-processing order.

The core bank contains brightness, contrast, and gamma operations in a photometric family, and Gaussian blur and additive Gaussian noise in a blur/noise family. Opposite directions, such as brightening and darkening, are separate operations. Each operation maps levels $m \in \{1, \ldots, M\}$ to prespecified physical parameters; zero denotes identity. The mappings, noise distribution, intensity clipping, and taper width are fixed after training-image review. Coordinates remain unchanged, but the edit can obscure damage or alter pavement context inside a box. Geometric warps

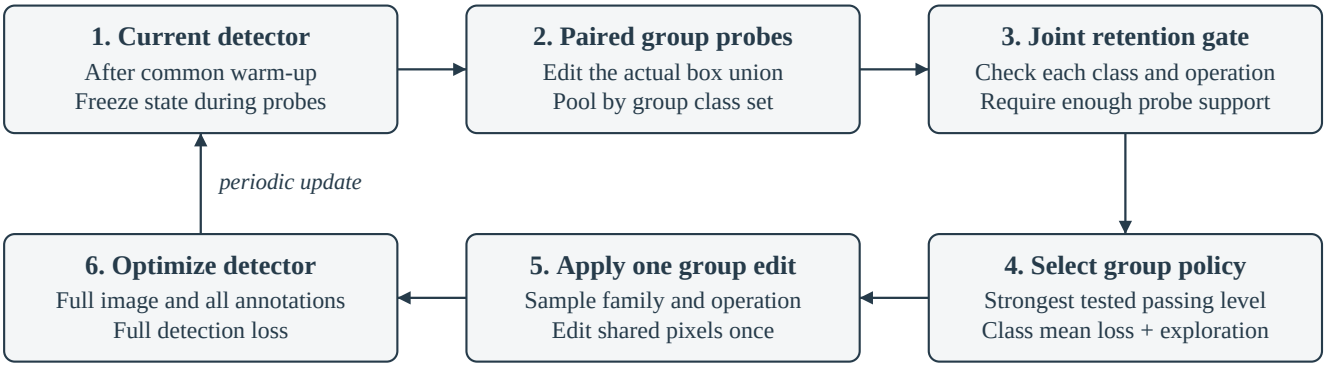


Fig. 1. Proposed training loop. Probes measure the actual union edit and pool class-specific feedback across groups with the same class set. Every represented class and operation must pass the empirical retention gate. Loss feedback weights eligible families; training uses the full detection loss. The controller is absent at inference.

and occlusion require additional annotation and visibility rules and are outside this core study.

D. Paired Detection Feedback

At each update, training images provide probe panels with one selected group per image. Within each class set, sampling follows the training image/group selection rule, using distinct source-image clusters where possible. The same panel evaluates the prespecified operation–strength grid. Only the selected group is edited; detector parameters, normalization statistics, preprocessing, and post-processing remain fixed. Noise uses a fixed number of seeded draws, shared across strength levels. These repeated draws improve measurement of stochastic effects but do not count as additional independent objects or clusters.

Let d_j^0 indicate a correct-class prediction matched to reference object j, using one-to-one matching at fixed confidence and intersection-over-union thresholds. Let $d_j^{o,m}$ denote the corresponding indicator after transformation, averaged over draws for a stochastic operation. For objects $\mathcal{P}_{s,c}$ of class c in selected groups with class set s, retention is

$$R_{c|s,o,m} = \frac{\sum_{j\in\mathcal{P}_{s,c}} d_j^0 d_j^{o,m}}{\sum_{j\in\mathcal{P}_{s,c}} d_j^0}. \tag{2}$$

The denominator counts reference-detected objects. A score is usable only with prespecified minimum numbers of such objects and contributing source-image clusters. Missing support is not a pass. Because this measure excludes reference misses, we also record unconditional recall, false positives, localization changes, and effects on objects outside the edited group. Unchanged pixels do not imply unchanged predictions elsewhere in the image.

E. Joint Strength Selection

A family is eligible at a strength only if every operation passes for every represented class. Let $\mathcal{O}_f$ contain the operations in family f, including their directions, and let $\mathcal{A}_{s,f}$ contain tested nonzero levels with adequate support satisfying

$$\min_{c\in s} \min_{o\in\mathcal{O}_f} R_{c|s,o,m} \geq \rho, \tag{3}$$

where ρ is the common retention threshold. The selected strength is

$$m_{s,f}^* = \max\big(\mathcal{A}_{s,f} \cup \{0\}\big). \tag{4}$$

Every activated level needs current probe evidence; passing a weaker level does not authorize an untested stronger one. Thus, a high average across classes cannot conceal a class-level retention failure. If no level passes, that family is unavailable for the current update.

For example, strong blur is rejected for crack–pothole groups if crack retention falls below ρ, even when pothole retention remains high. A weaker blur/noise level is eligible only if both blur and noise pass for both classes. Choosing the strongest passing level is a curriculum heuristic, not a proven optimum. The gate measures tolerance on finite probe panels; it neither guarantees retention for every group or noise realization nor establishes semantic preservation.

F. Class-Balanced Family Selection

Loss feedback weights only eligible families. We extract classification and localization losses for every annotated object in the selected group, using the same target-assignment procedure and loss weights in both views. Losses are averaged over predictions assigned to each object before averaging over objects. They are measured before confidence filtering, so missed detections still contribute supervised losses. The detector must expose these object-level terms; total image loss cannot substitute for them. Background and class-agnostic proposal losses remain in detector training but are excluded from family scoring.

Let $\bar{\ell}^{0}_{c|s}$ and $\bar{\ell}^{o,m}_{c|s}$ be mean losses over the same class-c objects, also averaging stochastic draws. The normalized class score is

$$A_{c|s,o,m} = \mathrm{clip}_{[0,B]}\left(\frac{\bar{\ell}^{o,m}_{c|s} - \bar{\ell}^{0}_{c|s}}{\bar{\ell}^{0}_{c|s} + \delta}\right), \tag{5}$$

with stabilizer $\delta > 0$ and cap $B > 0$. Averaging before clipping measures the net class-level increase; loss reductions are not discarded object by object. Normalization reduces loss-scale differences, and clipping bounds the score used for selection. At the selected strength,

$$D_{s,f} = \frac{1}{|s|\,|\mathcal{O}_f|}\sum_{c\in s}\sum_{o\in\mathcal{O}_f} A_{c|s,o,m^*_{s,f}}. \tag{6}$$

Each represented class has equal aggregation weight regardless of its box count. This does not imply equal performance gains, and a high score cannot override a failed retention gate.

For nonempty $\mathcal{E}_s = \{f : m^*_{s,f} > 0\}$, families are sampled using

$$p(f \mid s) = (1-\varepsilon)\frac{\exp(D_{s,f}/\tau)}{\sum_{h\in\mathcal{E}_s}\exp(D_{s,h}/\tau)} + \frac{\varepsilon}{|\mathcal{E}_s|}, \tag{7}$$

where $\tau > 0$ controls concentration and $0 < \varepsilon \le 1$ sets uniform exploration. An operation is sampled uniformly from the chosen family and applied at $m^*_{s,f}$; stochastic operations use a fresh draw from the probed distribution. An empty eligible set gives identity. Increased loss measures immediate difficulty, not future learning benefit. Setting $\varepsilon = 1$ tests uniform sampling as a simpler alternative.

G. Training Procedure and Evaluation Design

After warm-up, a fixed probability determines whether an image receives augmentation. One group is sampled uniformly, then edited using its class-set policy. Other groups can be selected on later visits. This restriction matches the single-group probe and avoids unmeasured interactions between simultaneous edits. It does not exclude overlapping groups. All annotations contribute to the full detection loss; images without target boxes receive no local edit and retain background supervision. We record group selection, actual nonidentity edits, and per-class exposure because augmentation opportunities vary with group count and policy eligibility.

Training and policy probes use only the training partition. Validation supports development and checkpoint selection; final test images remain unaugmented and never update the model or policy. Partitions respect known source-image clusters, without implying geographic independence. Comparisons hold the detector, initialization, preprocessing, data order, optimization budget, and checkpoint rule fixed within paired seeds. The approved operation bank and augmentation probability are shared; tuning budgets and total probe/search costs are reported separately.

The primary comparison is the full method against a validation-selected fixed shared image-level policy. To identify the source of any gain, a fixed-policy image/local pair changes only the edit region, and a shared/local versus class-conditioned/local pair changes policy conditioning while retaining the same grouping and feedback machinery. Further comparisons replace online control with a fixed class-conditioned policy, family-specific strengths with a common strength, or loss-weighted selection with uniform selection. The overlap ablation skips a selected multi-box group without resampling a substitute. This tests the value of treating overlaps; it does not keep the number of edits constant.

Evaluation will characterize class co-occurrence, group sizes, and probe sensitivity by class, source, and box size. Probe sensitivity alone cannot show that an augmentation helps or harms learning: matched fixed-policy runs will measure per-class AP changes relative to an identity-augmentation control on unaugmented images. The primary metric is mean AP across four classes and IoU thresholds 0.50:0.05:0.95, supplemented by AP50, per-class AP, and results on mixed-class and overlapping scenes. Paired seed differences, source-cluster uncertainty, fallback rates, exposure, and total training cost will reveal class-specific losses or costs hidden by aggregate gains.

The detector and numerical settings will be locked after a development pilot that verifies object-loss extraction, operation validity, and adequate probe support. Finite-panel selection can fluctuate; large connected groups may approach an image-level compromise, and rare class sets may often receive identity. The existing crop-classification baseline does not validate this detector study. The proposed contributions require evidence that local class-conditioned control and its feedback components improve detection beyond the matched simpler policies.

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